Structured questions

|!|BCELA0020000001|!| ★★ **SI**

A student carried out an investigation to find the glucose concentrations of three types of fruit juice. First, he performed the Benedict's test on glucose solutions of different concentrations. He then measured the absorbance of each solution with a colorimeter. (A colorimeter measures the amount of light that is absorbed by a coloured solution.) The results are recorded in the table below.

Concentration of glucose solution (M)	**Absorbance (arbitrary unit)**
0.1	0.20
0.2	0.40
0.3	0.55
0.4	0.75
0.5	0.90
0.6	1.00

(a) State ***two*** precautions that the student should have taken to ensure that the results give a valid comparison between the different glucose solutions. (2 marks)

(b) Using the graph paper on the last page of the test paper, plot a graph to show the results. (4 marks)

(c) Describe the relationship between the concentration of glucose solution and the absorbance of the solution based on the results. (1 mark)

(d) The student performed the Benedict's test on the three types of fruit juice. Explain how the student would use the graph you plotted in (b) to estimate the glucose concentration of the fruit juices. (2 marks)

##

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(a) Any ***two*** of the following: (2)

- Use the same volume of Benedict's solution.
- Use the same concentration of Benedict's solution.
- Boil the reaction mixture for the same length of time.

(b) Correct title, e.g. The absorbance of different concentrations of glucose solution (1)
Correct choice of axes (1)
Axes with labels and units (1)
Correct plotting and joining of line (1)

(c) The higher the glucose concentration, the higher the absorbance. (1)

(d) Measure the absorbance of the fruit juices with the same colorimeter. (1)
Find the concentration that corresponds to the absorbance from the graph. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Presenting information	4
(c)	Analysing data	1
(d)	Applying concepts	2

|!|BCELB0020000002|!| ★ **Cross topic**

The table below shows the concentration of some proteins, essential amino acids and fats in human milk.

Substance		Concentration / mg 100 cm^{-3}
Proteins	casein	140
	lactalbumin	218
	antibodies	364
	lipase	10
Essential amino acids	histidine	57
	isoleucine	221
	lysine	163
	methionine	33
Lipids	saturated fats	2000
	polyunsaturated fats	500
	monounsaturated fats	1700

(a) Describe how to test for the presence of proteins in human milk. (2 marks)

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(b) Based on the data provided, explain why human milk is suitable to be the main food source for infants. (4 marks)

(c) Lipase found in human milk becomes functional when it is stimulated by bile salts. Explain the advantage of human milk containing bile salt stimulated lipase. (3 marks)

##

(a) Dip a protein test strip into the human milk sample. (1)
The test end of the test strip gives a distinctive colour change if proteins are present. (1)

(b) Any ***two sets*** of the following: (2 + 2)

- Human milk is rich in proteins. (1)
  Proteins are essential for rapid growth / repair / formation of new cells. (1)
- Infants usually have weaker immunity. (1)
  Human milk contains antibodies, which help infants fight against pathogens. (1)
- Human milk contains essential amino acids, (1)
  which cannot be synthesized by infants' bodies. (1)

(c) When human milk is ingested and reaches the duodenum, it is mixed with bile. (1)
Bile contains bile salts. (1)
This can ensure that lipase becomes functional only when the milk reaches the small intestine which is the main digestion site of lipids. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Practical skills	2
(b)	Analysing data	2
	Applying concepts	2
(c)	Applying concepts	3

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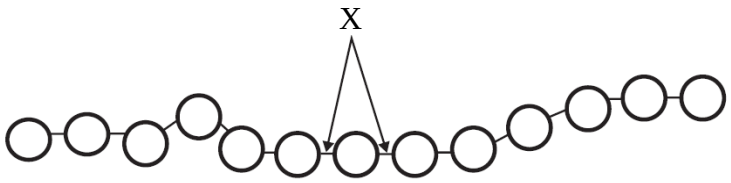
|!|BCELA0020000003|!| ★★ **Cross topic** **STSE**

CCEA GCE AS Biology Assessment Unit AS 1 Jan 2014 Q6

Proteins can be described as organic macromolecules.

(a) Explain the term 'organic macromolecule'. (2 marks)

(b) Gluten is a type of protein found in grains such as wheat, barley and rye. It is a 'composite' protein, made up of two simpler proteins called glutenin and gliadin. The following diagram represents the primary structure of part of a gliadin molecule.

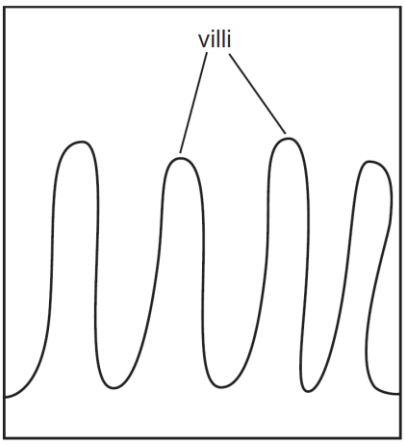


(i) Identify the building blocks of proteins, as represented by circles in the diagram. (1 mark)

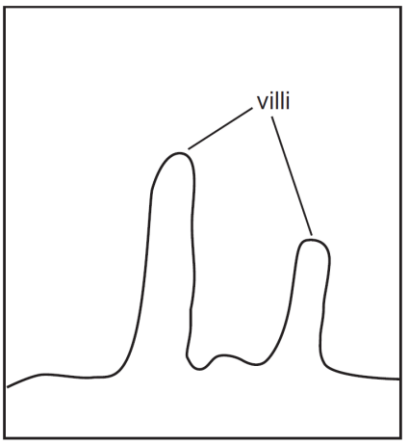
(ii) Identify the type of bond labelled X. (1 mark)

(c) Some people are intolerant of gluten in the diet. This is the basis of the condition known as coeliac disease. One of the proteins in gluten, gliadin, triggers an immune response in sufferers which results in damage to the lining of the ileum. This damage is often patchy and does not affect the whole ileum.
Sections of the ileum mucosal layer from a normal individual and from a coeliac sufferer are represented in the diagram below.

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normal



coeliac sufferer

(i) Describe the effect of coeliac disease on the structure of the ileum. (1 mark)

(ii) Coeliac sufferers are often nutrient-deficient. With reference to the damage caused to the ileum, suggest an explanation for this. (2 marks)

(iii) When investigating the ileum of a potential sufferer for signs of coeliac disease, several biopsies (tissue samples) are taken, each from a different region of the ileum. Suggest why. (1 mark)

##

(a) A large molecule consisting of subunits (1)
including carbon (and hydrogen) /originating in living organisms. (1)

(b) (i) amino acids (1)
(ii) peptide bond (1)

(c) (i) Villi are flattened / destroyed. (1)
(ii) The surface area of the ileum is reduced (1)
resulting in less effective absorption of nutrients. (1)
(iii) To increase the chances of finding the symptoms (1)

(8 marks)

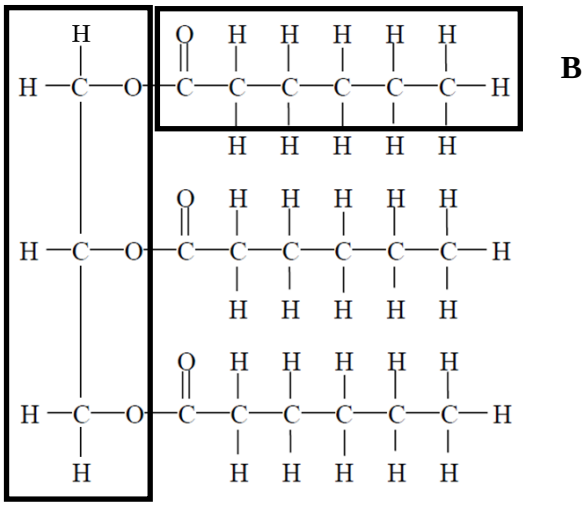
##

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	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Analysing data	1
(b)(ii)	Understanding basic concepts	1
(c)(i)	Analysing data	1
(c)(ii)	Applying concepts	2
(c)(iii)	Applying concepts	1

|!|BCELC0020000004|!|

The diagram below shows the structure of a lipid molecule.



(a) Name the parts labelled A and B. (2 marks)

(b) (i) Name this type of lipid. (1 mark)

(ii) State ***one*** function of this type of lipid. (1 mark)

(c) (i) Name the type of lipid molecule if B is replaced by a phosphate group. (1 mark)

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(ii) The lipid molecule in (c) (i) is an essential component of an important structure of living cells. Name this structure. (1 mark)

(d) Describe how you would test for the presence of lipids in a food sample. (4 marks)

##

(a) A is glycerol. (1)
B is fatty acid. (1)

(b) (i) triglyceride (1)
(ii) It acts as energy reserves / heat insulator / shock-absorber / waterproof layer to reduce water loss. (1)

(c) (i) phospholipids (1)
(ii) cell membrane (1)

(d) Add a small drop of the sample on a piece of filter paper. (1)
Leave the piece of filter paper in the air for about 10 minutes and a translucent spot is observed. (1)
Put the filter paper into an organic solvent. Take it out and let it dry. (1)
The disappearance of the translucent spot indicates the presence of lipids. (1)

(10 marks)

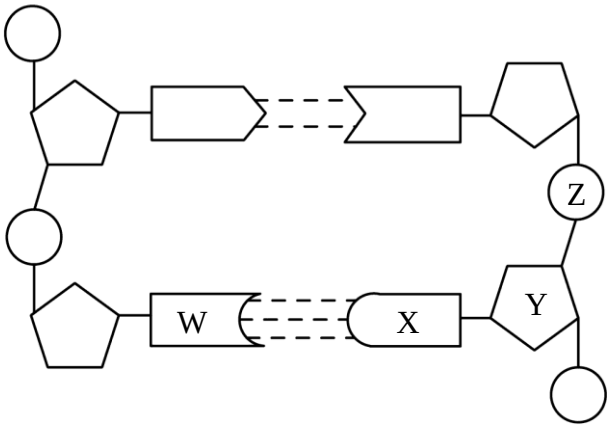
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)(i)	Understanding basic concepts	1
(b)(ii)	Understanding basic concepts	1
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	1
(d)	Practical skills	4

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|!|BCELA0020000005|!| ★★

The following figure shows a part of a DNA molecule.



(a) (i) Name Y and Z. (2 marks)

(ii) State the importance of the bond between W and X in DNA structure. (2 marks)

(b) In the 1950s, Erwin Chargaff determined the relative quantities of the four bases in DNA in different organisms. His results provided important evidence of the model of DNA proposed by James Watson and Francis Crick in 1953. Some of the Chargaff's data is shown in **Table I**.

Organism	Percentage of adenine	Percentage of thymine	Percentage of guanine	Percentage of cytosine
Chicken	28.0	28.4	22.0	21.6
Sea urchin	32.8	32.1	17.7	17.3
Octopus	33.2	31.6	17.6	17.6
Human	29.3	30.0	20.7	20.0

Table I

Describe the patterns shown by the data given in **Table I**. (3 marks)

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(c) **Table II** shows Chargaff's data for a virus.

Specimen	Percentage of adenine	Percentage of thymine	Percentage of guanine	Percentage of cytosine
Virus	24.0	31.2	23.3	21.5

Table II

(i) State how the result for the virus differs from the results of organisms given in **Table I**. (1 mark)

(ii) Suggest why the results for the virus are different from all the other organisms. (1 mark)

##

(a) (i) Y: deoxyribose sugar (1)
Z: phosphate (1)
(ii) Hydrogen bonds hold (two) polynucleotides strands chains together / holds complementary nucleotides together / prevents unwinding of strands. (1)
This gives stability to the DNA / makes DNA long lasting. (1)

(b) The percentage of adenine is almost equal to the percentage of thymine. (1)
The percentage of guanine is almost equal to the percentage of cytosine. (1)
The percentages of the four bases are different in different organisms. (1)

(c) (i) Percentage of adenine and thymine is not the same; percentage of guanine and cytosine is not the same. (1)
(ii) This virus may contain single-strand RNA. (1)

(9 marks)

##

	Skills assessed	Marks
(a)(i)	Analysing data	2
(a)(ii)	Applying concepts	2
(b)	Analysing data	3
(c)(i)	Analysing data	1
(c)(ii)	Applying concepts	1

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|!|BCELA0020000006|!| ★★ **Cross topic**

Sucrose is formed from glucose and another monosaccharide X by condensation. Sucrose can be broken down into glucose and monosaccharide X with the use of an enzyme Y.

$$\text{sucrose} + \text{water} \xrightarrow{\text{enzyme Y}} \text{glucose} + \text{monosaccharide X}$$

(a) Name monosaccharide X and enzyme Y. (2 marks)

(b) Name the reaction that breaks down sucrose into monosaccharides. (1 mark)

(c) When Benedict's solution is heated with solutions of sucrose, glucose and monosaccharide X respectively, what would be the results of the three sugar solutions? (3 marks)

(d) Sucralose is an artificial sweetener. It is synthesized from sucrose by replacing three hydroxyl group (OH) with chlorine atoms (Cl).
Figure I and **Figure II** show the molecules of sucrose and sucralose respectively.

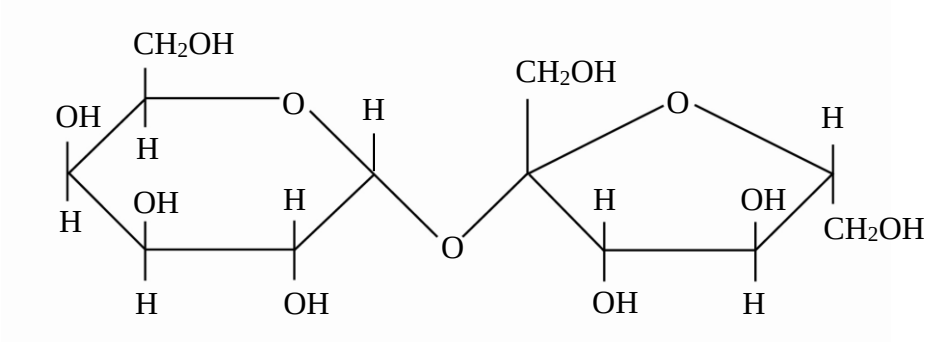


Figure I

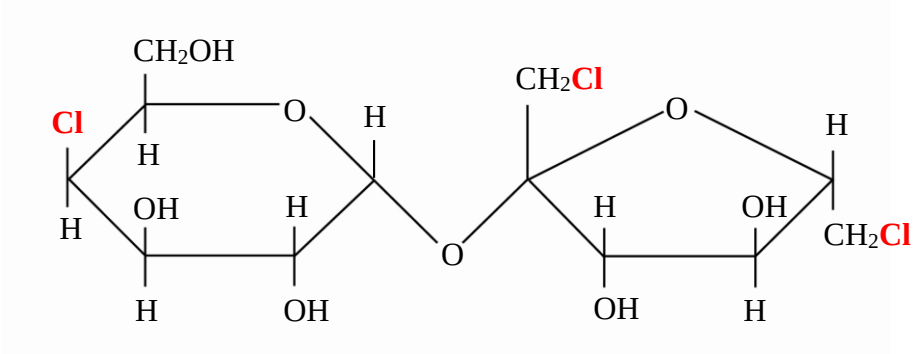


Figure II

Sucralose cannot be digested by enzyme Y and there are no other enzymes that can digest sucralose in the human body. Therefore, the consumption of this artificial sweetener does not lead to weight gain.

With reference to **Figure II**, explain why sucralose cannot be digested by enzyme Y. (3 marks)

##

(a) Monosaccharide X: fructose (1)

Enzyme Y: sucrase / invertase (1)

(b) hydrolysis (1)

(c) Sucrose: The solution remains blue. (1)

Glucose: A brick-red precipitate is formed. (1)

Monosaccharide X: A brick-red precipitate is formed. (1)

(d) The replacement of hydroxyl groups with chloride atoms gives a different molecular shape of sucralose when compared with sucrose. (1)

Enzyme Y is an enzyme that has specific shape / has an active site that is complementary to the shape of the substrate (i.e. sucrose). (1)

Sucrase and sucralose cannot form an enzyme-substrate complex and sucralose cannot be digested. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	1
(c)	Applying concepts	3
(d)	Applying concepts	3

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|!|BCELA0020000007|!| ★★ **SI**

The carbohydrates in green, unripe bananas are mostly in the form of starch. As the banana ages and ripens, the starch is converted to glucose, which gives the ripe fruit a sweet taste.

(a) What is the function of starch in unripe bananas? (1 mark)

(b) Describe how you would test for the presence of starch in a ripe banana. (2 marks)

(c) Suggest the steps that you would carry out to show that a ripe banana contains more reducing sugars than an unripe banana. (3 marks)

(d) Explain what happens to the water potential of a banana as it ripens. (2 marks)

(e) Suggest ***one*** benefit for the starch to be converted to glucose in the ripe bananas. (1 mark)

##

(a) Starch is the storage form of carbohydrates in bananas. (1)

(b) Add iodine solution to the banana sample. (1)
It turns blue-black in the presence of starch. (1)

(c) Prepare extracts of a ripe banana and an unripe bananas. (1)
Add Benedict's solution to the banana extracts and heat the mixtures. (1)
The more reducing sugars in the extract, the more brick-red precipitates are formed. (1)

(d) The water potential of the ripening banana decreases (1)
because more sugars are present in the banana and the sugars dissolve in water. (1)

(e) The sugary bananas can attract animals to eat them. This helps the seed dispersal of banana plants. (1)

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(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Practical skills	2
(c)	Practical skills	3
(d)	Applying concepts	2
(e)	Applying concepts	1

|!|BCELB0020000008|!| ★

In a laboratory, there are two bottles of organic chemicals (A and B). A label was found on each bottle, only stating the chemical formula as shown below.

	A	**B**
Chemical formula	$C_{12}H_{22}O_{11}$	$C_{16}H_{32}O_2$

(a) Calculate the hydrogen to oxygen ratios for chemicals A and B. (2 marks)

(b) Chemical A is a product of starch hydrolysis. What is chemical A? (1 mark)

(c) Chemical B belongs to a type of biomolecule.

(i) State the type of biomolecule to which chemical B belongs. (1 mark)

(ii) Describe how you would carry out a test to confirm that chemical B belongs to the biomolecule you answered in (i). (3 marks)

##

(a) For A, hydrogen to oxygen ratio = 22/11 : 1
= 2:1 (1)

For B, hydrogen to oxygen ratio = 32/2 : 1
= 16:1 (1)

(b) maltose (1)

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(c) (i) lipid (1)

(ii) Add a drop of chemical B onto a piece of filter paper and this will leave a translucent spot. (1)

Dip the filter paper into an organic solvent. (1)

Remove the paper from the solvent and the translucent spot will disappear after the filter paper is dried. (1)

(7 marks)

##

	Skills assessed	Marks
(a)	Mathematical skills	2
(b)	Applying concepts	1
(c)(i)	Analysing data	1
(c)(ii)	Practical skills	3

|!|BCELB0020000009|!| ★

The following table shows the nutrient content of low-fat yoghurt and high-fat yoghurt with equal mass.

Nutrient	Yoghurt type	
	Low-fat yoghurt	High-fat yoghurt
Lactose (g)	15.0	10.0
Protein (g)	12.0	9.0
Cholesterol (mg)	13.5	30.0
Total fat (g)	3.5	8.0

(a) Which type of yoghurt has greater energy content? (1 mark)

(b) Which two sugars is lactose composed of? (2 marks)

(c) Cholesterol is a kind of steroid and it is essential to humans.

(i) State ***two*** roles of cholesterol in living organisms. (2 marks)

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(ii) Explain how a diet rich in cholesterol might have a negative effect on the health of an individual. (3 marks)

##

(a) High-fat yoghurt (1)

(b) glucose (1)

galactose (1)

(c) (i) Any ***two*** of the following: (2)

- essential component of cell membrane
- making bile salts
- making steroid hormones
- making vitamin D

(Or other reasonable answers)

(ii) Cholesterol in the blood deposits on the inner wall of an artery. (1)

The lumen of artery gets narrower. (1)

This increases the risk of atherosclerosis and hence coronary heart disease / stroke. (1)

(8 marks)

##

	Skills assessed	Marks
(a)	Analysing data	1
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	2
(c)(ii)	Applying concepts	3

|!|BCELB0020000010|!| ★

A scientist analysed the organic molecules found in a bacterial cell. The following table shows the results of the analysis.

Organic molecule	Number of molecules per cell	Number of different kinds of molecules	Percentage of total dry mass of the cell
Glycogen	4 360	1	2.6
Lipid	22 000 000	4	9.2
Protein	2 360 000	1050	55.0
DNA	2	1	3.2
RNA	262 480	463	21.0

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(a) According to the table above, which organic molecule has the greatest molecular mass? Explain why. (Hint: The molecular mass of a substance is the mass of a molecule of that substance.) (2 marks)

(b) Explain why there is only one kind of glycogen molecule but there are many different kinds of protein molecules in the bacteria. (2 marks)

(c) Explain why it is necessary to have many different RNA molecules in the cell. (2 marks)

(d) The organic molecules constitute 91% of the dry mass of the bacterial cell. The remaining dry mass is constituted by inorganic substances like minerals.

(i) Suggest ***two*** minerals that are present in bacterial cells. (2 marks)

(ii) State ***one*** function of the two minerals in bacterial cells. (2 marks)

##

(a) DNA (1)
This is because each DNA molecule represents 1.6% of dry mass, which is the greatest among the organic molecules. (1)

(b) A glycogen molecule consists of only one kind of basic unit (i.e. glucose). (1)
A protein molecule is composed of different amino acids arranged in many different sequences. (1)

(c) Each amino acid combines with a specific tRNA in the translation process. (1)
Different mRNA codes for different proteins. (1)

(d) (i) sodium / nitrate / phosphate / other reasonable answer (Any ***two***, 2)

(ii) Any ***two*** of the following corresponding functions: (2)

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- Sodium is important for water balance of the cell.
- Nitrate is used in forming amino acids.
- Phosphate is the component of nucleic acids and cell membrane.

(Or other reasonable answer)

(10 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	2
(c)	Applying concepts	2
(d)(i)	Understanding basic concepts	2
(d)(ii)	Understanding basic concepts	2

|!|BCELA0020000011|!| ★★ **NOS**

Read the passage below and answer the following questions.

Chemical substances

Chemical substances are conventionally divided into two groups: organic and inorganic. Both are found in living organisms. It was previously thought that organic substances could only be made by living organisms, until the German chemist Friedrich Wöhler synthesized urea artificially in 1828. Since Wöhler's synthesis of urea, hundreds of organic compounds have been synthesized.

(a) What is the difference between organic and inorganic substances? (1 mark)

(b) Mineral is a kind of inorganic substance. Suggest ***one*** mineral and state its function in the human body. (2 marks)

(c) Urea can be formed from an organic substance in our bodies.

(i) State the organic substance. (1 mark)

(ii) State ***two*** functions of the organic substance. (2 marks)

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(iii) Briefly describe how urea is formed in the human body. (2 marks)

_____

_____

(d) Elaborate how Wöhler's synthesis of urea can be used to demonstrate the following aspect of the nature of science: Scientific knowledge is tentative and subject to change. (2 marks)

_____

_____

##

(a) All organic substances contain carbon atom(s) while inorganic substances have not (except carbon dioxide) (1)

(b) Any ***one*** example of mineral: (1)
iron / calcium / phosphate / other reasonable answer
Any ***one*** of the corresponding function: (1)
- Iron is needed for forming haemoglobin.
- Calcium is needed for muscle contraction / blood clotting.
- Phosphate is needed in forming nucleic acids / cell membrane.

(Or other reasonable answer)

(c) (i) protein (1)

(ii) Any ***two*** of the following: (2)
- It forms the cell membrane and cytoplasm of the cell.
- It forms enzyme / antibody / certain hormone.
- It provides energy if carbohydrates and lipids are used up.

(Or other reasonable answer)

(iii) Amino groups are removed from the amino acids in proteins. (1)
The amino groups are converted into urea. (1)

(d) Wöhler's synthesis of urea disproved the previous general belief that organic substances could only be made by living organisms. (1)
This shows that scientific knowledge will change when there is new evidence evolved. (1)

(10 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	2
(c)(i)	Understanding basic concepts	1
(c)(ii)	Understanding basic concepts	2
(c)(iii)	Understanding basic concepts	2

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(d)	Understanding basic concepts	2

|!|BCELA0020000012|!| ★★

Factor VIII is a glycoprotein found in blood plasma and is involved in blood clotting. This glycoprotein contains 2332 amino acids linked into a single polypeptide chain. This chain is then folded and coiled into a three-dimensional structure.

(a) What is a glycoprotein? (1 mark)

_____

(b) Besides Factor VIII, state ***one*** mineral that is essential in blood clotting. (1 mark)

_____

(c) Describe how amino acids are linked to form a single polypeptide chain of Factor VIII. (2 marks)

_____

_____

(d) Haemophilia A is a genetic disorder in which the sufferer is deficient of Factor VIII. It is caused by a change in the primary structure of Factor VIII. Explain how this change can lead to haemophilia. (3 marks)

_____

_____

_____

##

(a) A glycoprotein is a protein which has carbohydrate / sugar attached to it. (1)

(b) calcium (1)

(c) Amino acids are linked by condensation. (1)

Peptide bonds are formed between the amino group of one amino acid and the carboxyl group of another amino acid. (1)

(d) The sequence of amino acid changes. (1)

This causes the polypeptide chain to be coiled / foiled in a different way. (1)

The three-dimensional structure of Factor VIII changes and Factor VIII becomes non-functional. (1)

This causes haemophilia.

(7 marks)

##

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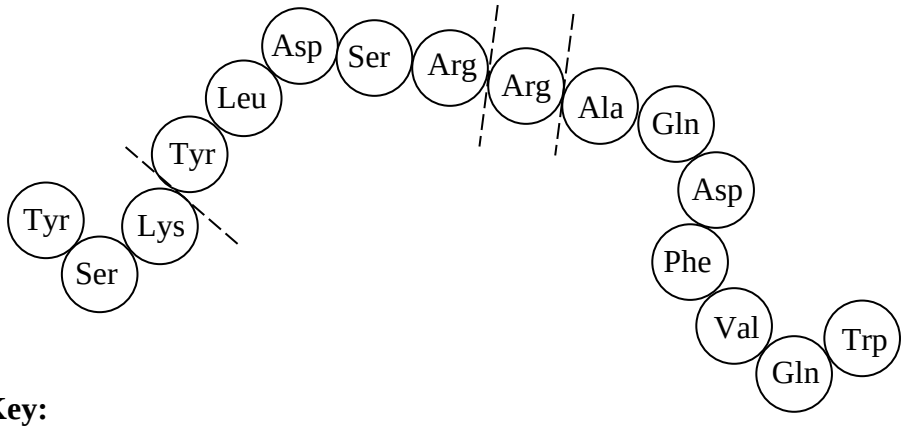
	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	2
(d)	Applying concepts	3

|!|BCELA0020000013|!| ★★

The first step in studying the primary structure of a protein is to break it into shorter lengths with enzymes. The following table shows some of the enzymes that can be used and the position of the peptide bonds they break.

Enzyme	Position of the peptide bond that enzyme breaks	
	First amino acid	Second amino acid
Trypsin	Lys or Arg	any
Pepsin	Phe,Trp or Tyr	any
Elastase	Gly, Ala or Val	any

The diagram below shows a polypeptide chain. The sequence of amino acids should be read from left to right. The dotted lines show the position of peptide bonds that are broken by trypsin.



Key:

– – – - Position of peptide bond that is broken by trypsin

(a) Water is essential for the reaction of breaking down the protein. State ***two*** significance of water in this reaction. (2 marks)

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(b) How many fragments will be produced from this polypeptide if it is incubated with a mixture of the enzyme pepsin? (1 mark)

(c) Explain why elastase and trypsin break peptide bonds between different amino acids. (3 marks)

(d) Pepsin is produced in the stomach while trypsin is produced in the pancreas. In the study of the primary structure of proteins, what pH of the medium should be used when pepsin and trypsin are used respectively? (2 marks)

##

(a) Any ***two*** of the following: (2)
- Water acts as a reactant in the hydrolysis process.
- Water provides a medium for the reaction to take place.
- Water provides a constant optimum temperature for the reaction.

(b) 4 (1)

(c) The active site of each enzyme has a specific shape. (1)
Only the substrate that fits exactly into the active site can bind to the enzyme (1)
to form the enzyme-substrate complex. (1)

(d) For pepsin: an acidic medium (1)
For trypsin: an alkaline medium (1)

(8 marks)

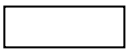
##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Analysing data	1
(c)	Applying concepts	3
(d)	Understanding basic concepts	2

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The following shows three components of a DNA molecule.

Phosphate group:  Deoxyibose:  Nitrogenous base: 

(a) Draw a diagram to show the structure of a DNA nucleotide using the symbols shown above. (1 mark)

(b) What is the shape of a DNA molecule? (1 mark)

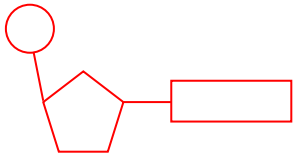
(c) Describe how DNA nucleotides are arranged in DNA molecules. (4 marks)

(d) Give ***two*** differences between RNA and DNA. (2 marks)

##

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(a)



(1)

(b) DNA molecule has a shape of a double helix. (1)

(c) The phosphate group of one nucleotide molecule combines with the deoxyribose of another. (1)

The sugar-phosphate chains form the backbones of the polynucleotide molecule. (1)

DNA consists of two polynucleotide strands joined by complementary base pairing. (1)

The base pairs in the two strands are held together by hydrogen bonds. (1)

(d) Any ***two*** of the following comparisons: (2)

RNA	DNA
• has a ribose sugar	• has a deoxyiribose sugar
• exists as a single strand	• exists as a double helix
• has the base uracil	• has the base thymine

(8 marks)

##

	Skills assessed	Marks
(a)	Presenting information	1
(b)	Understanding basic concepts	1
(c)	Applying concepts	4
(d)	Applying concepts	2

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A DNA nucleotide is composed of a deoxyribose, nitrogenous base and a substance X.

(a) Name substance X. (1 mark)

(b) Explain whether DNA is positively or negatively charged. (2 marks)

(c) A DNA fragment, consisting of 30 base pairs, was analysed for the number of different bases on each polynucleotide chain. The table below shows some of the results. Determine the missing values and complete the table. (2 marks)

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	Number of bases			
	Adenine	**Thymine**	**Guanine**	**Cytosine**
Chain 1	9		8	
Chain 2	6			

(d) Deoxyribose and glucose are both monosaccharides.

(i) State ***one*** molecular difference between deoxyribose and glucose molecules. (1 mark)

(ii) Describe how you can test whether deoxyribose is a reducing sugar. (3 marks)

##

(a) phosphate group (1)

(b) DNA is negatively charged (1)

due to the presence of the negatively charged phosphate group. (1)

(c) Deduct 1 mark for one wrong answer, get 0 mark for two or more wrong answers. (2)

	Number of bases			
	Adenine	**Thymine**	**Guanine**	**Cytosine**
Chain 1	9	**6**	8	**7**
Chain 2	6	**9**	**7**	**8**

(d) (i) Deoxyribose is a five-carbon sugar while glucose is a six-carbon sugar. (1)

(ii) Heat the deoxyribose solution with Benedict's solution. (1)

If a brick-red precipitate is formed, this shows that deoxyribose is a reducing sugar. (1)

If the Benedict's solution remains blue, this shows that deoxyribose is not a reducing sugar. (1)

(9 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Applying concepts	2
(c)	Analysing data	2
(d)(i)	Understanding basic concepts	1
(d)(ii)	Practical skills	3

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